

Experiment 13: V2X Latency Comparison

Objectives, MATLAB Programs and Output Images

Objective 1

To evaluate the End-to-End Latency (ms) of a V2X communication system under different vehicle operating conditions.

MATLAB Program

```
clc;
clear;
close all;

% Vehicle Speed (km/h)
speed = 20:20:120;

% Simulated End-to-End Latency (ms)
latency = [18 16 14 12 10 9];

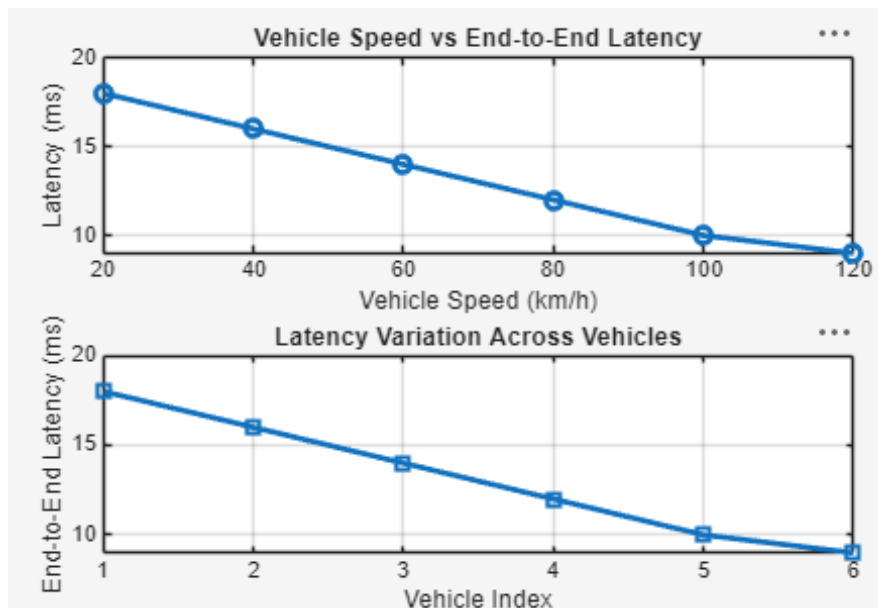
disp('Vehicle Speed (km/h)    End-to-End Latency (ms)')
disp([speed' latency'])

figure

% ----- Graph 1 -----
subplot(2,1,1)
plot(speed,latency,'-o','LineWidth',2)
title('Vehicle Speed vs End-to-End Latency')
xlabel('Vehicle Speed (km/h)')
ylabel('Latency (ms)')
grid on

% ----- Graph 2 -----
subplot(2,1,2)
plot(1:length(latency),latency,'-s','LineWidth',2)
title('Latency Variation Across Vehicles')
xlabel('Vehicle Index')
ylabel('End-to-End Latency (ms)')
grid on
```

Output Image – Objective 1



Objective 2

To analyze the Packet Delivery Ratio (%) and determine the reliability of V2X communication during data transmission.

MATLAB Program

```
clc;
clear;
close all;

% Vehicle Speed (km/h)
speed = 20:20:120;

% Simulated Packet Delivery Ratio (%)
PDR = [99 98 97 95 93 90];

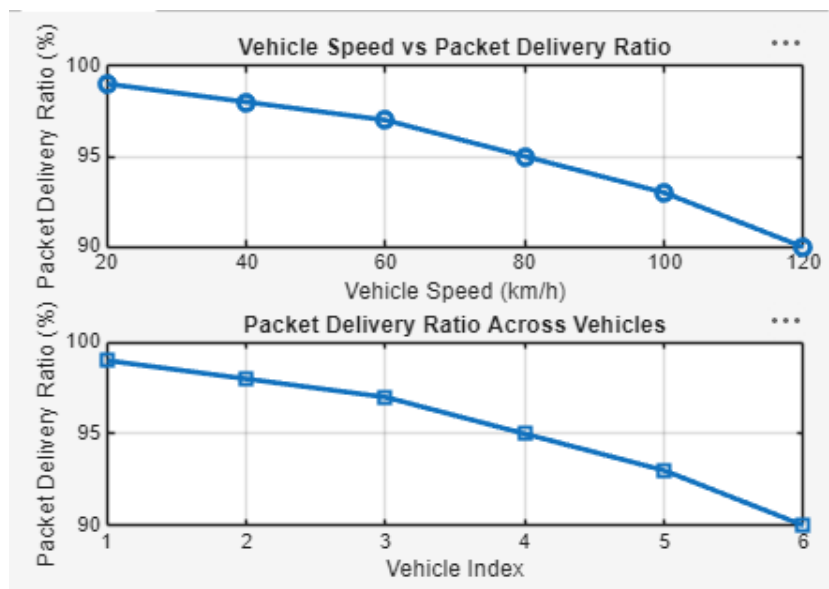
disp('Vehicle Speed (km/h)    Packet Delivery Ratio (%)')
disp([speed' PDR'])

figure

% ----- Graph 1 -----
subplot(2,1,1)
plot(speed,PDR,'-o','LineWidth',2)
title('Vehicle Speed vs Packet Delivery Ratio')
xlabel('Vehicle Speed (km/h)')
ylabel('Packet Delivery Ratio (%)')
grid on

% ----- Graph 2 -----
subplot(2,1,2)
plot(1:length(PDR),PDR,'-s','LineWidth',2)
title('Packet Delivery Ratio Across Vehicles')
xlabel('Vehicle Index')
ylabel('Packet Delivery Ratio (%)')
grid on
```

Output Image – Objective 2



Objective 3

To study the impact of Vehicle Speed (km/h) on the communication performance and stability of the V2X network.

MATLAB Program

```
clc;
clear;
close all;

% Vehicle IDs
vehicle = 1:6;

% Vehicle Speed (km/h)
speed = [30 45 60 75 90 105];

% Communication Range (m)
range = [280 260 240 220 200 180];

disp('Vehicle Speed(km/h) Communication Range(m)')
disp([vehicle' speed' range'])

figure

% ----- Graph 1 -----
subplot(2,1,1)
plot(vehicle,speed,'-o','LineWidth',2)
title('Vehicle Speed of Different Vehicles')
xlabel('Vehicle Index')
ylabel('Vehicle Speed (km/h)')
grid on

% ----- Graph 2 -----
subplot(2,1,2)
plot(speed,range,'-s','LineWidth',2)
title('Vehicle Speed vs Communication Range')
xlabel('Vehicle Speed (km/h)')
ylabel('Communication Range (m)')
grid on
```

Output Image – Objective 3

